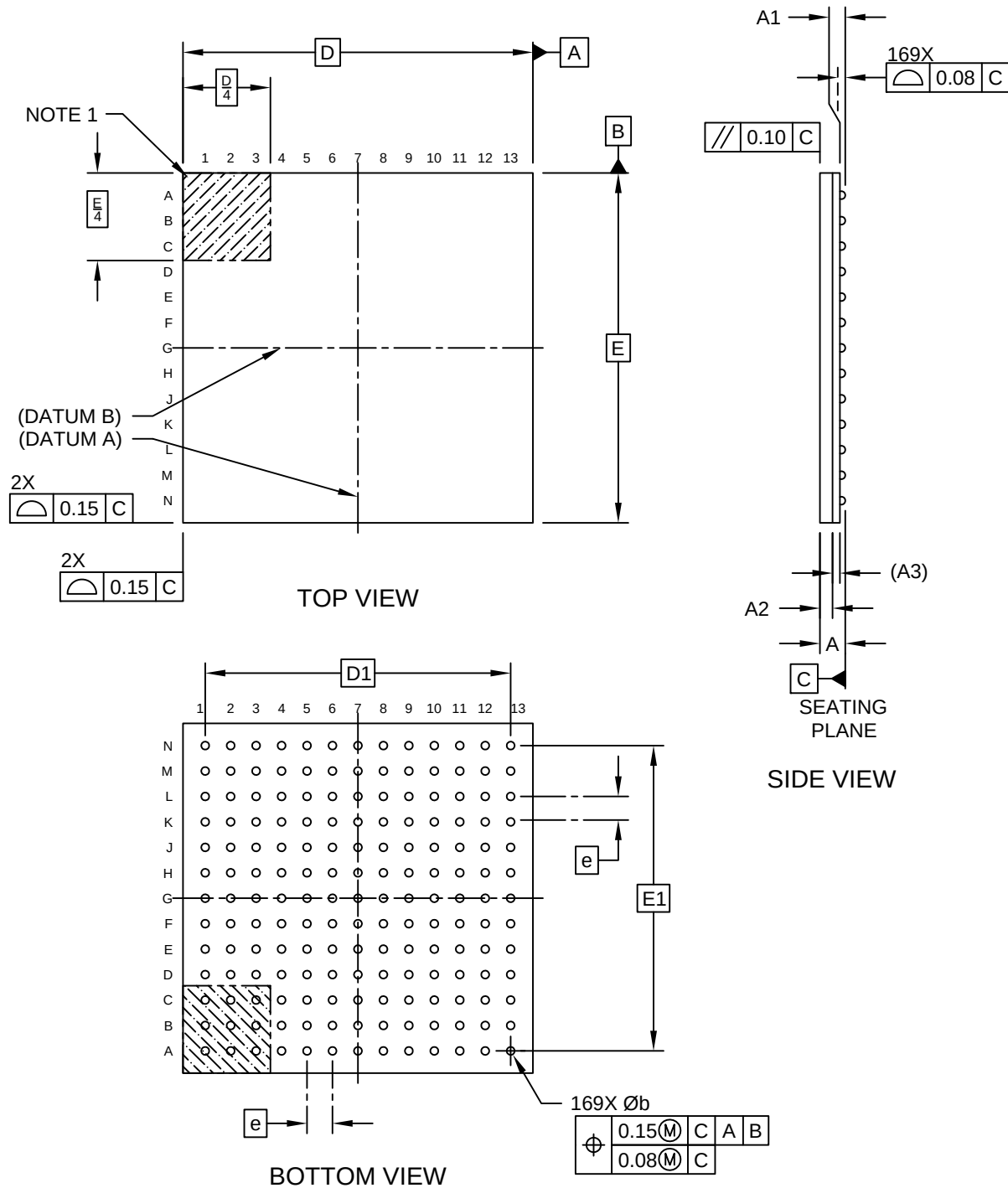


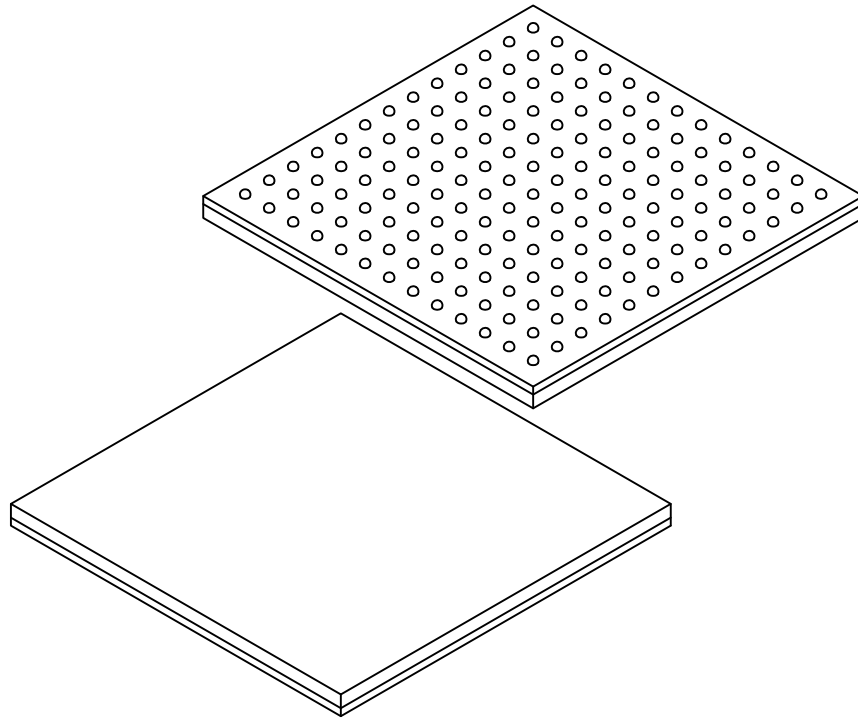
169-Ball Very, Very Thin Fine Pitch Ball Grid Array (TN) - 11x11 mm Body [WFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



169-Ball Very, Very Thin Fine Pitch Ball Grid Array (TN) - 11x11 mm Body [WFBGA]

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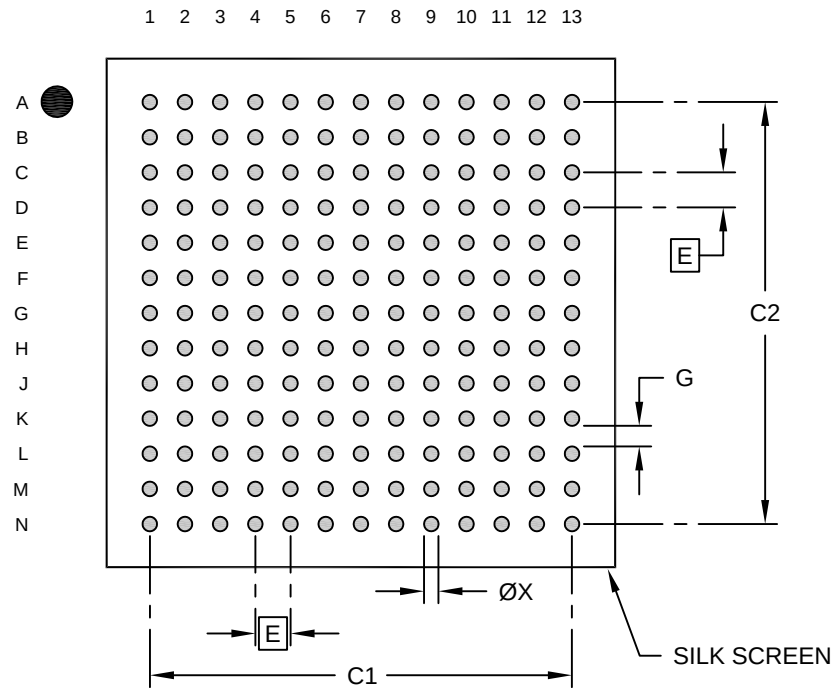
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	169		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	0.80
Standoff	A1	0.12	0.17	-
Mold Thickness	A2	0.35	0.40	0.45
Substrate Thickness	A3	0.26 REF		
Overall Length	D	11.00 BSC		
Overall Terminal Spacing	D1	9.60 BSC		
Overall Width	E	11.00 BSC		
Overall Terminal Spacing	E1	9.60 BSC		
Terminal Diameter	b	0.20	0.25	0.30

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

169-Ball Very, Very Thin Fine Pitch Ball Grid Array (TN) - 11x11 mm Body [WFBGA]

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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Overall Contact Pad Spacing	C1		9.60	
Overall Contact Pad Spacing	C2		9.60	
Contact Pad Width (X169)	X1			0.33
Contact Pad to Contact Pad	G	0.25		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.